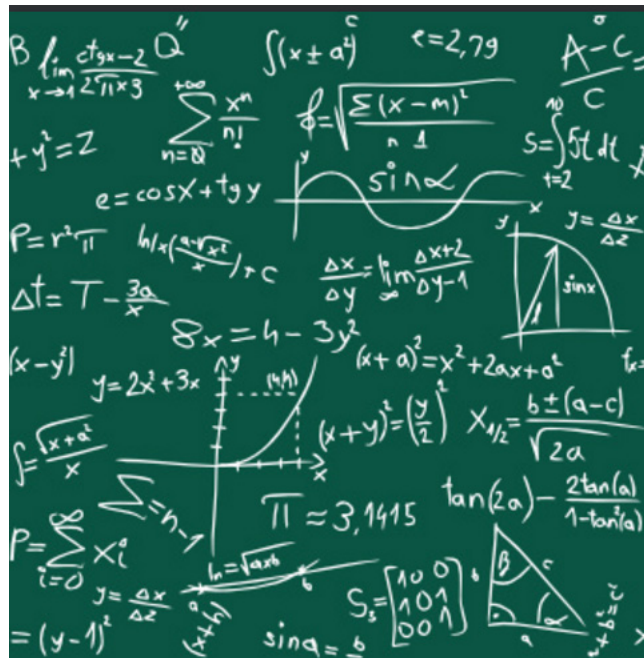


Exercises: limits of real functions of a real variable



- **Authors:** Fabio Furini and Gianluca Priori.
- **Book:** “Esercitazioni Analisi matematica 1”, Marco Bramanti, Esculapio.

We have the following fundamental limit:

$$\left(1 + \frac{\alpha}{\eta(x)}\right)^{\eta(x)} \rightarrow e^\alpha \quad \text{as } \eta(x) \rightarrow \pm\infty, \quad \alpha \in \mathbb{R}, \alpha \neq 0$$

We have the following 5 main fundamental limits:

$$\frac{\sin \varepsilon(x)}{\varepsilon(x)} \rightarrow 1 \quad \text{as } \varepsilon(x) \rightarrow 0$$

$$\frac{1 - \cos \varepsilon(x)}{\varepsilon^2(x)} \rightarrow \frac{1}{2} \quad \text{as } \varepsilon(x) \rightarrow 0$$

$$\frac{\log(1 + \varepsilon(x))}{\varepsilon(x)} \rightarrow 1 \quad \text{as } \varepsilon(x) \rightarrow 0$$

$$\frac{e^{\varepsilon(x)} - 1}{\varepsilon(x)} \rightarrow 1 \quad \text{as } \varepsilon(x) \rightarrow 0$$

$$\frac{(1 + \varepsilon(x))^\alpha - 1}{\varepsilon(x)} \rightarrow \alpha \quad \text{as } \varepsilon(x) \rightarrow 0, \quad \alpha \in \mathbb{R}, \alpha \neq 0$$

We also have the following fundamental limits of the hyperbolic functions:

$$\frac{\sinh \varepsilon(x)}{\varepsilon(x)} \rightarrow 1 \quad \text{as } \varepsilon(x) \rightarrow 0$$

$$\frac{\cosh \varepsilon(x) - 1}{\varepsilon^2(x)} \rightarrow \frac{1}{2} \quad \text{as } \varepsilon(x) \rightarrow 0$$

By the hierarchy of infinities theorem, we have:

$$\varepsilon^\beta(x) [\log_a \varepsilon(x)]^\alpha \rightarrow 0 \quad \text{as } \varepsilon(x) \rightarrow 0^+, \quad \alpha, \beta > 0, \quad a > 1$$

From the 5 main fundamental limits, the following asymptotic equivalences follow:

$$\sin \varepsilon(x) \sim \varepsilon(x) \quad \text{as } \varepsilon(x) \rightarrow 0$$

$$1 - \cos \varepsilon(x) \sim \frac{1}{2} \varepsilon^2(x) \quad \text{as } \varepsilon(x) \rightarrow 0$$

$$e^{\varepsilon(x)} - 1 \sim \varepsilon(x) \quad \text{as } \varepsilon(x) \rightarrow 0$$

$$\log(1 + \varepsilon(x)) \sim \varepsilon(x) \quad \text{as } \varepsilon(x) \rightarrow 0$$

$$(1 + \varepsilon(x))^\alpha - 1 \sim \alpha \varepsilon(x) \quad \text{as } \varepsilon(x) \rightarrow 0$$

From the 5 main fundamental limits, the following asymptotic expansions follow:

$$\sin \varepsilon(x) = \varepsilon(x) + o(\varepsilon(x)) \quad \text{as } \varepsilon(x) \rightarrow 0$$

$$\cos \varepsilon(x) = 1 - \frac{1}{2} \varepsilon^2(x) + o(\varepsilon^2(x)) \quad \text{as } \varepsilon(x) \rightarrow 0$$

$$\log(1 + \varepsilon(x)) = \varepsilon(x) + o(\varepsilon(x)) \quad \text{as } \varepsilon(x) \rightarrow 0$$

$$e^{\varepsilon(x)} = 1 + \varepsilon(x) + o(\varepsilon(x)) \quad \text{as } \varepsilon(x) \rightarrow 0$$

$$(1 + \varepsilon(x))^\alpha = 1 + \alpha \varepsilon(x) + o(\varepsilon(x)) \quad \text{as } \varepsilon(x) \rightarrow 0$$

From the fundamental limits, the following asymptotic equivalences also follow:

$$\sinh \varepsilon(x) \sim \varepsilon(x) \quad \text{as } \varepsilon(x) \rightarrow 0$$

$$\cosh \varepsilon(x) - 1 \sim \frac{1}{2} \varepsilon^2(x) \quad \text{as } \varepsilon(x) \rightarrow 0$$

For the hyperbolic functions we have the asymptotic expansions:

$$\sinh \varepsilon(x) = \varepsilon(x) + o(\varepsilon(x)) \quad \text{as } \varepsilon(x) \rightarrow 0$$

$$\cosh \varepsilon(x) = 1 + \frac{1}{2} \varepsilon^2(x) + o(\varepsilon^2(x)) \quad \text{as } \varepsilon(x) \rightarrow 0$$

Operations with “little o”:

$o(1)$ denotes a generic infinitesimal;

$$C \cdot o(x^n) = o(x^n);$$

$$x^m \cdot o(x^n) = o(x^{m+n});$$

$$o(x^m) \cdot o(x^n) = o(x^{m+n})$$

$$\frac{o(x^n)}{x^m} = o(x^{n-m}) \quad (n \geq m);$$

$$o(x^n) \pm o(x^m) = o(x^M) \quad \text{where } M = \min\{m, n\}.$$

Exercise 1

From $e^x = 1 + x + o(x)$, $x \rightarrow 0$, deduce that

$$\sinh(x) = x + o(x), \quad x \rightarrow 0.$$

For $x \rightarrow 0$, we have $e^{-x} = 1 - x + o(x)$, hence

$$\sinh x = \frac{e^x - e^{-x}}{2} = \frac{1}{2} \left((1 + x + o(x)) - (1 - x + o(x)) \right) = x + o(x).$$

Exercise 2

Using $\cosh^2 x - 1 = \sinh^2 x$ and the previous exercise, prove that

$$\lim_{x \rightarrow 0} \frac{\cosh x - 1}{x^2} = \frac{1}{2}$$

and deduce

$$\cosh x = 1 + \frac{1}{2} x^2 + o(x^2), \quad x \rightarrow 0.$$

From the previous exercise we have $\sinh^2 x \sim x^2$ as $x \rightarrow 0$ and moreover $\cosh 0 = 1$, hence

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\cosh x - 1}{x^2} \cdot \frac{\cosh x + 1}{\cosh x + 1} &= \lim_{x \rightarrow 0} \frac{\cosh^2 x - 1}{x^2} \cdot \underbrace{\lim_{x \rightarrow 0} \frac{1}{\cosh x + 1}}_{\rightarrow \frac{1}{2}} = \\ &= \frac{1}{2} \lim_{x \rightarrow 0} \frac{\sinh^2 x}{x^2} = \frac{1}{2} \lim_{x \rightarrow 0} \frac{x^2}{x^2} = \frac{1}{2}. \end{aligned}$$

Moreover

$$\left(\lim_{x \rightarrow 0} \frac{\cosh x - 1}{x^2} \right) - \frac{1}{2} = \lim_{x \rightarrow 0} \frac{\cosh x - 1 - \frac{1}{2}x^2}{x^2} = 0$$

which means

$$\begin{aligned} \cosh x - 1 - \frac{1}{2}x^2 &= o(x^2), \\ \cosh x &= 1 + \frac{1}{2}x^2 + o(x^2), \quad x \rightarrow 0. \end{aligned}$$

Exercise 3

Refine the equivalence

$$\log(x^2 + 1) \sim 2 \log x, \quad x \rightarrow +\infty$$

by proving that

$$\log(x^2 + 1) = 2 \log x + \frac{1}{x^2} + o\left(\frac{1}{x^2}\right), \quad x \rightarrow +\infty.$$

$$\log(x^2 + 1) = \log\left(x^2 \left(1 + \frac{1}{x^2}\right)\right) = \log x^2 + \log\left(1 + \frac{1}{x^2}\right).$$

Now

$$1/x^2 \rightarrow 0 \quad \text{as } x \rightarrow +\infty$$

and

$$\log(1 + y) = y + o(y) \quad \text{as } y \rightarrow 0,$$

hence

$$\log(x^2 + 1) = 2 \log x + \frac{1}{x^2} + o\left(\frac{1}{x^2}\right), \quad x \rightarrow +\infty.$$

Exercise 4

Compute the following limit of a function of a real variable:

$$\lim_{x \rightarrow +\infty} \log x - \sqrt{x}$$

$$\lim_{x \rightarrow +\infty} \log x - \sqrt{x} = \lim_{x \rightarrow +\infty} -\sqrt{x} = -\infty.$$

Exercise 5

Compute the following limit of a function of a real variable:

$$\lim_{x \rightarrow +\infty} 2^x - x^2$$

$$\lim_{x \rightarrow +\infty} 2^x - x^2 = \lim_{x \rightarrow +\infty} 2^x = +\infty.$$

Exercise 6

Compute the following limit of a function of a real variable:

$$\lim_{x \rightarrow +\infty} \frac{\log(x^2 + 1)}{2^x}$$

We use the equivalence $\log(x^2 + 1) \sim 2 \log x$ as $x \rightarrow +\infty$:

$$\lim_{x \rightarrow +\infty} \frac{\log(x^2 + 1)}{2^x} = \lim_{x \rightarrow +\infty} \frac{2 \log x}{2^x} = 0.$$

Exercise 7

Compute the following limit of a function of a real variable:

$$\lim_{x \rightarrow +\infty} \left(\frac{x+2}{x+1} \right)^x$$

We use the fundamental limit

$$\lim_{x \rightarrow \pm\infty} \left(1 + \frac{1}{x}\right)^x = e.$$

$$\lim_{x \rightarrow +\infty} \left(\frac{x+2}{x+1}\right)^x = \lim_{x \rightarrow +\infty} \left[\left(1 + \frac{1}{x+1}\right)^{x+1}\right]^{\frac{x}{x+1}} = e^1 = e.$$

Exercise 8

Compute the following limit of a function of a real variable:

$$\lim_{x \rightarrow 0^+} x^{\log x}$$

We have

$$x^{\log x} = e^{\log x^{\log x}} = e^{\log x \log x},$$

hence

$$\lim_{x \rightarrow 0^+} x^{\log x} = \lim_{x \rightarrow 0^+} e^{\overbrace{\log x}^{-\infty} \overbrace{\log x}^{-\infty}} = e^{+\infty} = +\infty.$$

Moreover, the base x tends to 0 and the exponent $\log x$ tends to $-\infty$, hence the limit is not an indeterminate form:

$$\lim_{x \rightarrow 0^+} x^{\log x} = 0^{-\infty} = +\infty$$

Exercise 9

Let

$$\lim_{x \rightarrow +\infty} x(\log(x+2) - \log x) = L.$$

- (a) $L = 2$
- (b) L does not exist
- (c) $L = 1$
- (d) None of the other answers is correct.

$$\lim_{x \rightarrow +\infty} x(\log(x+2) - \log x) = \lim_{x \rightarrow +\infty} x \log \left(1 + \frac{2}{x} \right) = \lim_{x \rightarrow +\infty} x \cdot \frac{2}{x} = 2.$$

The correct answer is (a).

Exercise 10

Compute the following limit of a quotient using equivalent power functions for each term:

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{\sin x}$$

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{\sin x} = \lim_{x \rightarrow 0} \frac{x}{x} = 1.$$

Exercise 11

Compute the following limit of a quotient using equivalent power functions for each term:

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin^2 x}$$

We have

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2} \text{ hence } 1 - \cos x \sim \frac{1}{2} x^2 \text{ as } x \rightarrow 0.$$

Moreover, we have

$$\sin x \sim x \text{ as } x \rightarrow 0 \text{ hence } (\sin x)^2 \sim x^2 \text{ as } x \rightarrow 0.$$

We then have:

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin^2 x} = \lim_{x \rightarrow 0} \frac{\frac{1}{2} x^2}{x^2} = \frac{1}{2}$$

Exercise 12

Compute the following limit of a quotient using equivalent power functions for each term:

$$\lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 4x}$$

$$\lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 4x} = \lim_{x \rightarrow 0} \frac{3x}{4x} = \frac{3}{4}$$

Exercise 13

Compute the following limit of a quotient using equivalent power functions for each term:

$$\lim_{x \rightarrow 0} \frac{(1 - e^{2x})^2}{1 - \cos 5x}$$

$$\lim_{x \rightarrow 0} \frac{(1 - e^{2x})^2}{1 - \cos 5x} = \lim_{x \rightarrow 0} \frac{(-2x)^2}{\frac{1}{2}(5x)^2} = \frac{8}{25}$$

Exercise 14

Compute the following limit of a quotient using equivalent power functions for each term:

$$\lim_{x \rightarrow 0} \frac{\sin x^3}{(1 - e^x)^3}$$

$$\lim_{x \rightarrow 0} \frac{\sin x^3}{(1 - e^x)^3} = \lim_{x \rightarrow 0} \frac{x^3}{(-x)^3} = -1.$$

Exercise 15

Compute

$$\lim_{x \rightarrow 0} x^2 e^{\frac{\sqrt{\pi}}{x} \sin(x \log 7)}$$

We analyze the exponent using the equivalence $\sin(x \log 7) \sim x \log 7$ as $x \rightarrow 0$:

$$\lim_{x \rightarrow 0} \frac{\sqrt{\pi}}{x} \sin(x \log 7) = \lim_{x \rightarrow 0} \frac{\sqrt{\pi}}{x} \cdot x \log 7 = \sqrt{\pi} \log 7.$$

The given limit equals

$$\lim_{x \rightarrow 0} x^2 e^{\frac{\sqrt{\pi}}{x} \sin(x \log 7)} = 0 \cdot e^{\sqrt{\pi} \log 7} = 0 \cdot 7^{\sqrt{\pi}} = 0.$$

Exercise 16

Compute

$$\lim_{x \rightarrow 0^+} \left(\frac{1}{\sin x} + \log x \right)$$

Using

$$\lim_{x \rightarrow 0^+} \sin x \log x = \lim_{x \rightarrow 0^+} x \log x = 0,$$

we have

$$\lim_{x \rightarrow 0^+} \left(\frac{1}{\sin x} + \log x \right) = \lim_{x \rightarrow 0^+} \frac{1 + \sin x \log x}{\sin x} = +\infty$$

since the numerator $1 + \sin x \log x$ tends to 1 while the denominator $\sin x$ is a positive function tending to 0.

Equivalently, again from

$$\lim_{x \rightarrow 0^+} \frac{\log x}{\frac{1}{\sin x}} = \lim_{x \rightarrow 0^+} \sin x \log x = \lim_{x \rightarrow 0^+} x \log x = 0,$$

we have that $1/\sin x$ is an infinity of higher order than $\log x$ as $x \rightarrow 0^+$, hence

$$\lim_{x \rightarrow 0^+} \left(\frac{1}{\sin x} + \log x \right) = \lim_{x \rightarrow 0^+} \frac{1}{\sin x} = +\infty.$$

We also prove that

$$\lim_{x \rightarrow 0^+} x \log x = 0$$

We make the change of variable $x = \frac{1}{y}$, hence

$$\lim_{x \rightarrow 0^+} x \log x = \lim_{y \rightarrow +\infty} \frac{1}{y} \log \frac{1}{y} = \lim_{y \rightarrow +\infty} \frac{-\log y}{y} = 0$$

Exercise 17

Compute

$$\lim_{x \rightarrow 0} \frac{(1 - \cos x)^3 - \sin x^6}{x^6}$$

From

$$1 - \cos x \sim \frac{1}{2}x^2, \sin x \sim x, \quad x \rightarrow 0,$$

we have

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{(1 - \cos x)^3 - \sin x^6}{x^6} &= \lim_{x \rightarrow 0} \frac{(1 - \cos x)^3}{x^6} - \lim_{x \rightarrow 0} \frac{\sin x^6}{x^6} = \\ &= \lim_{x \rightarrow 0} \frac{\frac{1}{8}x^6}{x^6} - \lim_{x \rightarrow 0} \frac{x^6}{x^6} = \frac{1}{8} - 1 = -\frac{7}{8}. \end{aligned}$$

Equivalently, using the calculus of Landau symbols, from

$$\cos x = 1 - \frac{1}{2}x^2 + o(x^2), \sin x = x + o(x), \quad x \rightarrow 0,$$

we have

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{(1 - \cos x)^3 - \sin x^6}{x^6} &= \lim_{x \rightarrow 0} \frac{\frac{1}{8}x^6 - x^6 + o(x^6)}{x^6} = \\ \lim_{x \rightarrow 0} \frac{-\frac{7}{8}x^6 + o(x^6)}{x^6} &= \lim_{x \rightarrow 0} \frac{-\frac{7}{8}x^6}{x^6} = -\frac{7}{8}. \end{aligned}$$

Exercise 18

Compute

$$\lim_{x \rightarrow 0} \frac{8(1 - \cos x)^3 - \sin x^6}{x^6}$$

From

$$1 - \cos x \sim \frac{1}{2}x^2, \sin x \sim x, \quad x \rightarrow 0,$$

we have

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{8(1 - \cos x)^3 - \sin x^6}{x^6} &= \lim_{x \rightarrow 0} \frac{8(1 - \cos x)^3}{x^6} - \lim_{x \rightarrow 0} \frac{\sin x^6}{x^6} = \\ &= \lim_{x \rightarrow 0} \frac{x^6}{x^6} - \lim_{x \rightarrow 0} \frac{x^6}{x^6} = 1 - 1 = 0. \end{aligned}$$

Equivalently, using the calculus of Landau symbols, from

$$\cos x = 1 - \frac{1}{2}x^2 + o(x^2), \sin x = x + o(x), \quad x \rightarrow 0,$$

we have

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{8(1 - \cos x)^3 - \sin x^6}{x^6} &= \lim_{x \rightarrow 0} \frac{x^6 - x^6 + o(x^6)}{x^6} = \\ \lim_{x \rightarrow 0} \frac{o(x^6)}{x^6} &= 0 \end{aligned}$$

since the numerator $o(x^6)$, by the very definition of the Landau symbol, is an infinitesimal of higher order than the denominator x^6 .

We have the asymptotic expansions as $x \rightarrow 0$:

$$\sin x = x + o(x), \cos x = 1 - \frac{1}{2}x^2 + o(x^2), e^x = 1 + x + o(x),$$

$$\sinh x = x + o(x), \cosh x = 1 + \frac{1}{2}x^2 + o(x^2).$$

Exercise 19

Compute the following limits using the comparison of infinitesimals

$$\lim_{x \rightarrow 0} \frac{(\sin x)^2 + x}{x^3 - \sin x}$$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{(\sin x)^2 + x}{x^3 - \sin x} &= \lim_{x \rightarrow 0} \frac{(x + o(x))^2 + x}{x^3 - x + o(x)} = \\ &= \lim_{x \rightarrow 0} \frac{x^2 + o(x^2) + x}{x^3 - x + o(x)} = \lim_{x \rightarrow 0} \frac{x + o(x)}{-x + o(x)} = \lim_{x \rightarrow 0} \frac{x}{-x} = -1. \end{aligned}$$

Exercise 20

Compute the following limits using the comparison of infinitesimals

$$\lim_{x \rightarrow 0} \frac{|1 - \cos x + \sin x|}{(e^x - 1)^2}$$

$$\lim_{x \rightarrow 0} \frac{|1 - \cos x + \sin x|}{(e^x - 1)^2} = \lim_{x \rightarrow 0} \frac{|\frac{1}{2}x^2 + o(x^2) + x + o(x)|}{(x + o(x))^2} =$$

$$\lim_{x \rightarrow 0} \frac{|x + o(x)|}{x^2 + o(x^2)} = \lim_{x \rightarrow 0} \frac{|x|}{x^2} = \lim_{x \rightarrow 0} \frac{1}{|x|} = +\infty.$$

Exercise 21

Compute the following limits using the comparison of infinitesimals

$$\lim_{x \rightarrow 0} \frac{\sin 2x - \sin^2 x}{e^x - 1}$$

$$\lim_{x \rightarrow 0} \frac{\sin 2x - \sin^2 x}{e^x - 1} = \lim_{x \rightarrow 0} \frac{2x + o(x) - (x + o(x))^2}{x + o(x)} =$$

$$= \lim_{x \rightarrow 0} \frac{2x + o(x) - x^2 + o(x^2)}{x + o(x)} = \lim_{x \rightarrow 0} \frac{2x + o(x)}{x + o(x)} = \lim_{x \rightarrow 0} \frac{2x}{x} = 2.$$

Exercise 22

Compute the following limits using the comparison of infinitesimals

$$\lim_{x \rightarrow 0} \frac{\sin^3 x - \sin 4x}{\cosh x - 1 + x}$$

$$\lim_{x \rightarrow 0} \frac{\sin^3 x - \sin 4x}{\cosh x - 1 + x} = \lim_{x \rightarrow 0} \frac{(x + o(x))^3 - 4x + o(x)}{\frac{1}{2}x^2 + o(x^2) + x} =$$

$$= \lim_{x \rightarrow 0} \frac{x^3 + o(x^3) - 4x + o(x)}{x + o(x)} = \lim_{x \rightarrow 0} \frac{-4x + o(x)}{x + o(x)} = \lim_{x \rightarrow 0} \frac{-4x}{x} = -4.$$

Exercise 23

Compute the following limits using the comparison of infinitesimals

$$\lim_{x \rightarrow 0} \frac{\sin x - \sin x^2}{\sinh(2x) - \sinh(2x)^2}$$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin x - \sin x^2}{\sinh(2x) - \sinh(2x)^2} &= \lim_{x \rightarrow 0} \frac{x + o(x) - x^2 + o(x^2)}{2x + o(x) - (2x + o(x))^2} = \\ &= \lim_{x \rightarrow 0} \frac{x + o(x)}{2x + o(x) - 4x^2 + o(x^2)} = \lim_{x \rightarrow 0} \frac{x + o(x)}{2x + o(x)} = \lim_{x \rightarrow 0} \frac{x}{2x} = \frac{1}{2}. \end{aligned}$$

Exercise 24

Compute

$$\lim_{x \rightarrow 1^+} \frac{\log(1 + \sqrt{x-1})}{\sqrt{x^2-1}}$$

We have

$$\lim_{x \rightarrow 1^+} \frac{\log(1 + \sqrt{x-1})}{\sqrt{x^2-1}} = \lim_{x \rightarrow 1^+} \frac{\log(1 + \sqrt{x-1})}{\sqrt{x-1}\sqrt{x+1}} = \frac{1}{\sqrt{2}} \lim_{x \rightarrow 1^+} \frac{\log(1 + \sqrt{x-1})}{\sqrt{x-1}}.$$

Setting $y = \sqrt{x-1}$, the limit becomes

$$\frac{1}{\sqrt{2}} \lim_{y \rightarrow 0^+} \frac{\log(1+y)}{y} = \frac{1}{\sqrt{2}}$$

by the fundamental limit

$$\lim_{y \rightarrow 0} \frac{\log(1+y)}{y} = 1.$$

Exercise 25

Compute

$$\lim_{x \rightarrow +\infty} \left(\frac{x^2 + 3x}{x^2 - 5x} \right)^{(x + \log x)}$$

We have

$$\begin{aligned}\lim_{x \rightarrow +\infty} \left(\frac{x^2 + 3x}{x^2 - 5x} \right)^{(x+\log x)} &= \lim_{x \rightarrow +\infty} \left(\frac{x+3}{x-5} \right)^{(x+\log x)} = \lim_{x \rightarrow +\infty} \left(1 + \frac{8}{x-5} \right)^{(x+\log x)} \\ &= \lim_{x \rightarrow +\infty} \left(1 + \frac{1}{\frac{x-5}{8}} \right)^{(x+\log x)}\end{aligned}$$

Using the fundamental limit

$$\lim_{x \rightarrow \pm\infty} \left(1 + \frac{1}{x} \right)^x = e$$

we obtain

$$\lim_{x \rightarrow +\infty} \left(\frac{x^2 + 3x}{x^2 - 5x} \right)^{(x+\log x)} = \lim_{x \rightarrow +\infty} \left[\left(1 + \frac{1}{\frac{x-5}{8}} \right)^{\frac{x-5}{8}} \right]^{\frac{8(x+\log x)}{x-5}} = e^8$$

since

$$\frac{8(x+\log x)}{x-5} \sim \frac{8x}{x-5} \xrightarrow{x \rightarrow +\infty} 8.$$

Exercise 26

Compute

$$\lim_{x \rightarrow 0} \frac{x(\pi^x - e^x)}{\cos(x) - 1}$$

We have

$$\lim_{x \rightarrow 0} \frac{x(\pi^x - e^x)}{\cos(x) - 1} = \lim_{x \rightarrow 0} \frac{x\pi^x \left(1 - \left(\frac{e}{\pi} \right)^x \right)}{\cos(x) - 1} = \lim_{x \rightarrow 0} \pi^x \frac{x \left(1 - \left(\frac{e}{\pi} \right)^x \right)}{\cos(x) - 1}$$

Since

$$1 - \left(\frac{e}{\pi} \right)^x \sim -x \log \left(\frac{e}{\pi} \right)$$

and also

$$\cos(x) - 1 \sim -\frac{1}{2}x^2$$

we obtain

$$\lim_{x \rightarrow 0} \frac{x(\pi^x - e^x)}{\cos(x) - 1} = \lim_{x \rightarrow 0} \pi^x \frac{-x^2 \log \left(\frac{e}{\pi} \right)}{-\frac{1}{2}x^2} = 2 \log \left(\frac{e}{\pi} \right).$$

Exercise 27

Compute

$$\lim_{x \rightarrow 0} \frac{\sin(\ln(1-x))}{1-2^x}$$

The limit is in fact immediate if we observe that, as $x \rightarrow 0$,

$$\ln(1-x) \sim -x$$

and also

$$1-2^x \sim -x \log 2$$

thus obtaining

$$\lim_{x \rightarrow 0} \frac{\sin(\ln(1-x))}{1-2^x} = \lim_{x \rightarrow 0} \frac{\sin(-x)}{-x \log 2} = \frac{1}{\log 2}$$

since

$$\sin(-x) \sim -x$$

Exercise 28

Compute

$$\lim_{x \rightarrow +\infty} \frac{x^4 \sin^2(\pi - 2 \arctan(x))}{x^2 + 3}$$

Recalling that

$$\arctan(x) \xrightarrow{x \rightarrow +\infty} \frac{\pi}{2}$$

and hence that

$$\pi - 2 \arctan(x) \xrightarrow{x \rightarrow +\infty} 0$$

we can immediately use the asymptotic estimate

$$\sin^2(\pi - 2 \arctan(x)) \sim (\pi - 2 \arctan(x))^2$$

as well as

$$\frac{x^4}{x^2 + 3} \sim x^2$$

and say that

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{x^4 \sin^2(\pi - 2 \arctan(x))}{x^2 + 3} &= \lim_{x \rightarrow +\infty} \frac{x^4 (\pi - 2 \arctan(x))^2}{x^2 + 3} \\ &= \lim_{x \rightarrow +\infty} x^2 (\pi - 2 \arctan(x))^2 \end{aligned}$$

which is still an indeterminate form of the type $[0 \cdot +\infty]$. However, with the change of variable $y = \pi - 2 \arctan(x)$ and observing that

$$x = \tan\left(\frac{\pi}{2} - \frac{y}{2}\right) = \cot\left(\frac{y}{2}\right)$$

we obtain

$$\begin{aligned} \lim_{x \rightarrow +\infty} x^2 (\pi - 2 \arctan(x))^2 &= \lim_{y \rightarrow 0} y^2 \cot^2\left(\frac{y}{2}\right) \\ &= \lim_{y \rightarrow 0} \frac{y^2}{\sin^2\left(\frac{y}{2}\right)} \cos^2\left(\frac{y}{2}\right) = 4. \end{aligned}$$

Exercise 29

Compute

$$\lim_{x \rightarrow 1} \frac{\log(x^x) - \log x}{1 - \cos(x - 1)}$$

Recalling that

$$\log(x^x) - \log x = \log(e^{x \log x}) - \log x = x \log x - \log x = (x - 1) \log x$$

we immediately have

$$\lim_{x \rightarrow 1} \frac{\log(x^x) - \log x}{1 - \cos(x - 1)} = \lim_{x \rightarrow 1} \frac{(x - 1) \log x}{1 - \cos(x - 1)}$$

Setting, for greater clarity in the use of the asymptotic estimates, $x - 1 = t$, we obtain by substitution

$$\lim_{x \rightarrow 1} \frac{(x - 1) \log x}{1 - \cos(x - 1)} = \lim_{t \rightarrow 0} \frac{t \log(1 + t)}{1 - \cos t} = 2$$

since

$$\frac{t \log(1 + t)}{1 - \cos t} \sim \frac{t^2}{\frac{1}{2}t^2}$$

Exercise 30

Compute

$$\lim_{x \rightarrow +\infty} x \log \left(\frac{3x + x^2}{1 + x + x^2} \right)$$

We immediately observe that the limit is in the indeterminate form of the type $[+\infty \cdot 0]$, since

$$\frac{3x + x^2}{1 + x + x^2} \xrightarrow{x \rightarrow +\infty} 1$$

In order to resolve the indeterminate form, we can add and subtract 1 inside the argument of the logarithm:

$$\log\left(\frac{3x + x^2}{1 + x + x^2}\right) = \log\left(1 + \frac{3x + x^2}{1 + x + x^2} - 1\right)$$

At this point, since

$$\frac{3x + x^2}{1 + x + x^2} - 1 \xrightarrow{x \rightarrow +\infty} 0$$

we can use the well-known asymptotic estimate of the logarithm, valid for some $\varepsilon(x) \rightarrow 0$, obtaining

$$\log\left(1 + \frac{3x + x^2}{1 + x + x^2} - 1\right) \sim \frac{3x + x^2}{1 + x + x^2} - 1 = \frac{2x - 1}{1 + x + x^2}$$

The limit reduces to

$$\lim_{x \rightarrow +\infty} x \left(\frac{2x - 1}{1 + x + x^2} \right) = 2.$$

Exercise 31

Compute

$$\lim_{x \rightarrow +\infty} \left(e^{\sqrt{x^2 + x}} - e^{\sqrt{x^2 - 1}} \right)$$

Factoring out $e^{\sqrt{x^2-1}}$ we obtain

$$\lim_{x \rightarrow +\infty} e^{\sqrt{x^2-1}} \left(e^{\sqrt{x^2+x}-\sqrt{x^2-1}} - 1 \right).$$

We examine the indeterminate form $\sqrt{x^2+x}-\sqrt{x^2-1}$ as $x \rightarrow +\infty$:

$$\begin{aligned} \lim_{x \rightarrow +\infty} \sqrt{x^2+x}-\sqrt{x^2-1} &= \lim_{x \rightarrow +\infty} \frac{\sqrt{x^2+x}-\sqrt{x^2-1}}{\frac{\sqrt{x^2+x}-\sqrt{x^2-1}}{\sqrt{x^2+x}+\sqrt{x^2-1}}} = \lim_{x \rightarrow +\infty} \frac{x^2+x-x^2+1}{\sqrt{x^2+x}+\sqrt{x^2-1}} \\ &= \lim_{x \rightarrow +\infty} \frac{x+1}{x\left(\sqrt{1+\frac{1}{x}}+\sqrt{1-\frac{1}{x^2}}\right)} = \frac{1}{2}. \end{aligned}$$

Hence the given limit equals

$$\lim_{x \rightarrow +\infty} e^{\sqrt{x^2-1}} \left(e^{\sqrt{x^2+x}-\sqrt{x^2-1}} - 1 \right) = (\sqrt{e}-1) \lim_{x \rightarrow +\infty} e^{\sqrt{x^2-1}} = +\infty.$$

Equivalently, from $1/x \rightarrow 0$ as $x \rightarrow +\infty$ and from

$$\sqrt{1+y} = 1 + (1/2)y + o(y)$$

as $y \rightarrow 0$, we have

$$\sqrt{x^2+x} = x\sqrt{1+\frac{1}{x}} = x\left(1 + \frac{1}{2x} + o\left(\frac{1}{x}\right)\right) = x + \frac{1}{2} + o(1), \quad x \rightarrow +\infty,$$

where $o(1)$ denotes a generic infinitesimal, and

$$\sqrt{x^2-1} = x\sqrt{1-\frac{1}{x^2}} = x\left(1 - \frac{1}{2x^2} + o\left(\frac{1}{x^2}\right)\right) = x - \frac{1}{2x} + o\left(\frac{1}{x}\right), \quad x \rightarrow +\infty.$$

Hence the given limit equals

$$\lim_{x \rightarrow +\infty} \left(e^{\sqrt{x^2+x}} - e^{\sqrt{x^2-1}} \right) = \lim_{x \rightarrow +\infty} \left(e^{x+1/2+o(1)} - e^{x-1/(2x)+o(1/x)} \right) =$$

$$\lim_{x \rightarrow +\infty} e^x \left(e^{1/2+o(1)} - e^{-1/(2x)+o(1/x)} \right) = (\sqrt{e}-1) \lim_{x \rightarrow +\infty} e^x = +\infty.$$

We have that $1/x \rightarrow 0$ as $x \rightarrow +\infty$ and we have the asymptotic expansions as $y \rightarrow 0$

$$\sin y = y + o(y), \quad \cos y = 1 - \frac{1}{2}y^2 + o(y^2), \quad e^y = 1 + y + o(y), \quad \log(1+y) = y + o(y).$$

Exercise 32

Compute the following limit

$$\lim_{x \rightarrow +\infty} \frac{1 + x^4 \sin(1/x^4)}{x^2(1 - \cos(1/x^2))}$$

From the asymptotic expansion we have:

$$\cos\left(\frac{1}{x^2}\right) = 1 - \frac{1}{2} \left(\frac{1}{x^2}\right)^2 + o\left(\left(\frac{1}{x^2}\right)^2\right) = 1 - \frac{1}{2} \frac{1}{x^4} + o\left(\frac{1}{x^4}\right) \text{ as } x \rightarrow +\infty.$$

Expanding the ratio asymptotically, we have:

- for the numerator

$$1 + x^4 \sin\left(\frac{1}{x^4}\right) = 1 + x^4 \left(\frac{1}{x^4} + o\left(\frac{1}{x^4}\right)\right) = 1 + 1 + o(1) \text{ as } x \rightarrow +\infty;$$

- for the denominator

$$\begin{aligned} x^2 \left[1 - \cos\left(\frac{1}{x^2}\right)\right] &= x^2 \left[1 - \left(1 - \frac{1}{2} \frac{1}{x^4} + o\left(\frac{1}{x^4}\right)\right)\right] = \\ &= x^2 \left[\frac{1}{2} \frac{1}{x^4} - o\left(\frac{1}{x^4}\right)\right] = \frac{1}{2} \frac{1}{x^2} - o\left(\frac{1}{x^2}\right) = \frac{1}{2} \frac{1}{x^2} + o\left(\frac{1}{x^2}\right) \text{ as } x \rightarrow +\infty. \end{aligned}$$

Hence we obtain:

$$\lim_{x \rightarrow +\infty} \frac{1 + x^4 \sin\left(\frac{1}{x^4}\right)}{x^2 \left[1 - \cos\left(\frac{1}{x^2}\right)\right]} = \lim_{x \rightarrow +\infty} \frac{2 + o(1)}{\frac{1}{2} \frac{1}{x^2} + o\left(\frac{1}{x^2}\right)} = +\infty$$

where, as always, $o(1)$ denotes a generic infinitesimal.

Exercise 33

Compute the following limit

$$\lim_{x \rightarrow +\infty} \frac{x^2 e^{-1/x^4} - x^2}{\log(x^2 + 1) - 2 \log x}$$

$$\begin{aligned}\lim_{x \rightarrow +\infty} \frac{x^2(e^{-1/x^4} - 1)}{\log(x^2 + 1) - \log x^2} &= \lim_{x \rightarrow +\infty} \frac{x^2\left(-\frac{1}{x^4} + o\left(\frac{1}{x^4}\right)\right)}{\log\left(1 + \frac{1}{x^2}\right)} = \\ &= \lim_{x \rightarrow +\infty} \frac{-\frac{1}{x^2} + o\left(\frac{1}{x^2}\right)}{\frac{1}{x^2} + o\left(\frac{1}{x^2}\right)} = \lim_{x \rightarrow +\infty} \frac{-\frac{1}{x^2} + o\left(\frac{1}{x^2}\right)}{\frac{1}{x^2} + o\left(\frac{1}{x^2}\right)} = \lim_{x \rightarrow +\infty} \frac{-\frac{1}{x^2}}{\frac{1}{x^2}} = -1.\end{aligned}$$

Exercise 34

Compute

$$\lim_{x \rightarrow 0} \frac{e^{-x^4} - x^4 - 1}{x^2 \cos x - x^2}$$

We use the expansion

$$e^{-x^4} = 1 - x^4 + o(x^4) \text{ as } x \rightarrow 0$$

and the equivalence

$$\cos x - 1 \sim -x^2/2 \text{ as } x \rightarrow 0.$$

Hence we have

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{e^{-x^4} - x^4 - 1}{x^2(\cos x - 1)} &= \lim_{x \rightarrow 0} \frac{1 - x^4 + o(x^4) - x^4 - 1}{x^2\left(-\frac{x^2}{2}\right)} = \\ &= \lim_{x \rightarrow 0} \frac{-2x^4 + o(x^4)}{-\frac{x^4}{2}} = \lim_{x \rightarrow 0} \frac{-2x^4}{-\frac{x^4}{2}} = 4.\end{aligned}$$

Exercise 35

Compute

$$\lim_{x \rightarrow \pi/4} (2 \sin^2 x)^{\frac{1}{\cos 2x}}$$

Using $\cos 2x = 1 - 2\sin^2 x$ and setting $y = 1 - 2\sin^2 x$, we observe that $y \rightarrow 0$ as $x \rightarrow \pi/4$ and the given limit equals

$$\lim_{y \rightarrow 0} (1 - y)^{\frac{1}{y}} = \lim_{y \rightarrow 0} e^{\frac{\log(1-y)}{y}} = e^{-1} = \frac{1}{e}$$

also by virtue of the fundamental limit

$$\lim_{y \rightarrow 0} \frac{\log(1 - y)}{y} = -1.$$

Exercise 36

Compute

$$\lim_{x \rightarrow 0^+} \frac{(1 + x^5)^{\frac{1}{x^2 \sin 2x}} - 1}{2 \log(1 + x^3)}$$

We write the numerator in the form

$$e^{\frac{\log(1+x^5)}{x^2 \sin 2x}} - 1$$

and we analyze the exponent. As $x \rightarrow 0$ we have

$$\frac{\log(1 + x^5)}{x^2 \sin 2x} \sim \frac{x^5}{x^2 \cdot 2x} = \frac{x^2}{2}.$$

From this and from $e^y - 1 \sim y$ as $y \rightarrow 0$, it follows that

$$e^{\frac{\log(1+x^5)}{x^2 \sin 2x}} - 1 \sim \frac{\log(1 + x^5)}{x^2 \sin 2x} \sim \frac{x^2}{2}.$$

The given limit equals

$$\lim_{x \rightarrow 0^+} \frac{(1 + x^5)^{\frac{1}{x^2 \sin 2x}} - 1}{2 \log(1 + x^3)} = \lim_{x \rightarrow 0^+} \frac{\frac{x^2}{2}}{2x^3} = \frac{1}{4} \lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty.$$

Exercise 37

Compute

$$\lim_{x \rightarrow 0^+} \frac{1 - x^x}{x^2}$$

Writing the numerator in the form

$$1 - e^{x \log x},$$

the exponent $x \log x$ tends to zero as $x \rightarrow 0^+$. Since $1 - e^y \sim -y$ as $y \rightarrow 0$, it follows that

$$1 - e^{x \log x} \sim -x \log x, \quad x \rightarrow 0^+.$$

The given limit equals

$$\lim_{x \rightarrow 0^+} \frac{1 - x^x}{x^2} = \lim_{x \rightarrow 0^+} \frac{-x \log x}{x^2} = \lim_{x \rightarrow 0^+} \frac{-\log x}{x} = +\infty.$$

Exercise 38

Compute

$$\lim_{x \rightarrow 0} \frac{\sin(e^{x^3} - 1)}{\sqrt{1 + x^3} - 1}$$

We have

$$\sin(e^{x^3} - 1) \sim e^{x^3} - 1 \sim x^3, \quad x \rightarrow 0^+$$

and

$$\sqrt{1 + x^3} - 1 \sim \frac{1}{2}x^3, \quad x \rightarrow 0^+.$$

The given limit equals

$$\lim_{x \rightarrow 0} \frac{\sin(e^{x^3} - 1)}{\sqrt{1 + x^3} - 1} = \lim_{x \rightarrow 0} \frac{x^3}{\frac{1}{2}x^3} = 2.$$

Exercise 39

Compute

$$\lim_{x \rightarrow 0^+} \frac{x^x - e^x + x^\alpha}{\sin \sqrt{x}} = L$$

as the parameter $\alpha > 0$ varies.

We have

$$\lim_{x \rightarrow 0^+} \frac{x \log x}{x^\alpha} = \begin{cases} 0, & \alpha < 1 \\ -\infty, & \alpha = 1 \end{cases}$$

hence, as $x \rightarrow 0^+$, the infinitesimal $x \log x$ is of lower order than x ($x = o(x \log x)$) but of higher order than x^α if $\alpha < 1$ ($x \log x = o(x^\alpha)$, $\alpha < 1$).

Hence, from the expansion

$$e^y = 1 + y + o(y), \quad y \rightarrow 0,$$

it follows

$$x^x - e^x + x^\alpha = e^{x \log x} - e^x + x^\alpha =$$

$$1 + x \log x + o(x \log x) - 1 - x + o(x) + x^\alpha =$$

$$\begin{cases} x \log x + o(x \log x), & \alpha \geq 1 \\ x^\alpha + o(x^\alpha), & 0 < \alpha < 1. \end{cases}$$

The given limit equals

$$\lim_{x \rightarrow 0^+} \frac{x^x - e^x + x^\alpha}{\sin \sqrt{x}} = \begin{cases} \lim_{x \rightarrow 0^+} \frac{x \log x}{\sqrt{x}} = \lim_{x \rightarrow 0^+} \sqrt{x} \log x = 0, & \alpha \geq 1 \\ \lim_{x \rightarrow 0^+} \frac{x^\alpha}{\sqrt{x}} = 0, & 1/2 < \alpha < 1 \\ \lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{\sqrt{x}} = 1, & \alpha = 1/2 \\ \lim_{x \rightarrow 0^+} \frac{x^\alpha}{\sqrt{x}} = +\infty, & 0 < \alpha < 1/2. \end{cases}$$